TYPICAL UNIT 3 CALCULATION MCQ'S IN ME 265 - DETAILED SOLUTIONS

1. Thermal efficiency of heat engine $=$ network $\frac{\text{output}}{\text{heat}}$ input

$\eta_{TH} = \frac{30}{80} \times 100\ \% = 0.375 \times 100\ \% = \mathbf{37.5}\ \%$

2. $COP_R = \frac{\dot{Q}_L}{\dot{W}_{in}} = \frac{(360/60)}{2} = \mathbf{3}$

3. $COP_{hp} = \frac{\dot{Q}_L}{\dot{W}_{in}}$; $\dot{W}_{in} = \frac{Q}{COP} = \frac{80000\ \text{kJ/h}}{2.5} = \mathbf{32,000\ kJ/h}$

4. $Q = mC_p\Delta T = 1 \times 1.005(35 - 15) = 20.1\ \text{kW}$

$COP_R = \frac{T_L}{T_H - T_L} = \frac{288.15}{308.15 - 288.15} = 14.4075$

Power needed $W = \frac{20.1}{14.4075} = \mathbf{1.40\ kW}$

5. $COP_{hp} = \frac{T_H}{T_H - T_L} = \frac{293.15}{286.84} = 1.022$

Power required $= \dot{W}_{in} = \frac{Q}{COP} = \frac{100}{1.022} = \mathbf{97.85\ kW}$

6. $\dot{W}_{net} = \dot{Q}_H - \dot{Q}_L = 80 - 50 = \mathbf{30\ mW}$

7. $\dot{Q}_{in} = \frac{\dot{W}_{net}}{\eta_{TH}} = \frac{50}{0.24} = 208.3\ \text{kW}$

Fuel consumption rate, $\dot{m}_f = \frac{\dot{Q}_{in}}{CV} = \frac{208.3\ \text{kJ/s}}{44000\ \text{kJ/kg}} = \mathbf{0.00473\ kg/s}$

8. $\dot{Q}_L = 360\ \text{kJ/min}$

$\dot{W}_{in} = 2\frac{\text{kJ}}{\text{s}} = \frac{2\ \text{kJ}}{\left(\frac{1}{60}\right)\text{min}} = 120\ \text{kJ/min}$

$\dot{Q}_H = \dot{W}_{in} + \dot{Q}_L = 360 + 120 = \mathbf{480\ kJ/min}$

9. $\dot{Q}_H = 80{,}000\ \text{kJ/h}$

$COP = 2.5$; $\dot{W}_{in} = \frac{80000}{2.5} = 32{,}000\ \text{kJ/h}$

$\dot{Q}_L = \dot{Q}_H - \dot{W}_{in} = 80{,}000 - 32{,}000 = \mathbf{48,000\ kJ/h}$

10. $\eta_{Carnot} = 1 - \frac{400}{1000} = 60\ \%$

Actual efficiency $= \frac{300}{325} = 0.615 = 61.5\ \%$

$\eta_{actual} > \eta_{ideal}$, therefore, **this is impossible**